General Capabilities

Critical and creative thinking

In Technologies, students develop critical and creative thinking as they imagine, generate, iterate and critically evaluate ideas. Students analyse problems, refine concepts and reflect on the decision-making process by engaging in systems, design and computational thinking.

Students think critically and creatively while considering how data, information, systems, materials, tools and equipment (past and present) impact on our lives, and how these elements might be better designed and managed. Experimenting, drawing, modelling, designing and working with equipment and software helps students to build their visual and spatial thinking and to create solutions, products, services and environments.

Digital literacy

Technologies gives students the opportunity to become discerning users, productive creators, critical analysts and effective developers of digital solutions. Development of digital literacy allows students to operate and manage digital systems and practise digital safety and wellbeing while investigating, creating and communicating. As students develop digital literacy skills, they build their understanding of how to utilise digital tools when designing digital solutions. Students learn how to operate specific digital tools to help them realise their design ideas. This may occur when investigating needs or opportunities or researching and analysing information. It also occurs when students generate and communicate design ideas, processes and solutions, and when they develop plans, schedules and processes and collaborate online to create innovative and enterprising solutions.

Ethical understanding

Technologies develops students' capacity to understand and apply ethical and socially responsible principles when managing projects and collaborating with others and creating, sharing and using technologies. Using an ethical lens, they investigate past, current and future local, national, regional and global technological priorities. Understanding the protection of data, intellectual property and individual privacy helps students to be respectful creators.
Students explore complex issues and consider possibilities and ethical implications. They learn about safe and ethical procedures, and consider the rights of others and their responsibilities in using sustainable practices. Students learn to appreciate and value the part they play in the social and natural systems they live in.

Literacy

In Technologies, students develop literacy by interpreting a wide range of practical texts including technical manuals and specifications. They evaluate content in diverse formats, analyse data and reports and navigate complex information. Students learn to use technical symbols, icons and terminology, adapting language for both general and specialised contexts. They produce purpose-driven texts such as annotated drawings, software guides and project plans tailored to specific needs and audiences.

Numeracy

In Technologies, students develop the capacity to interpret and use mathematical knowledge and skills in a range of real-life situations. They use numbers to calculate, measure and estimate; interpret and draw conclusions from statistics; measure and record throughout the process of generating and iterating ideas; develop, refine and test concepts; and cost and sequence when making products and managing projects. In using software, materials, tools and equipment, students work with the concepts of numbers, geometry, scale, proportion, measurement and volume. They use three-dimensional models, create accurate technical drawings, work with digital models and use computational thinking in decision-making processes when designing and creating best-fit solutions.

Personal and social capability

Students develop personal and social capabilities in Technologies through engaging in project management and design activities in a collaborative environment. They show initiative in their learning, apply design thinking skills and make informed decisions independently. Working in teams, they develop cooperation, leadership, conflict resolution and resource-sharing skills. Embracing innovation and risk fosters resilience as they navigate uncertainty.
Students also cultivate social responsibility by considering the impacts of decisions on people, communities and environments. They deepen empathy and respect by understanding diverse user needs and reflect on how digital tools and environments such as social media affect their wellbeing, applying appropriate strategies to both face-to-face and digital environments.

Pre-primary

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning builds on the Early Years Learning Framework and each child's funds of knowledge. A holistic curriculum that integrates knowledge,

understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Digital Technologies provides opportunities for children to explore digital systems and the use of digital technologies, while developing an understanding of what the internet is, participating in digital contexts safely and developing foundational skills in computational thinking.

In Pre-primary, children discover the uses of digital technologies in everyday life. They explore common objects and images that exist within data they encounter and use this data to make meaning. Children develop their understandings of digital safety and personal data through a variety of ways, such as teacher modelling or role-play. They learn the steps to take when encountering inappropriate content, pop-ups or uninitiated contact. Children identify problems and design solutions, such as following a sequence of steps (algorithm) to achieve an outcome.

Achievement Standard

By the end of the year:

Children identify and label digital systems (hardware and software), where they are used and their common features. They identify that data can be represented as objects or images. Children follow safety strategies in online encounters. They identify that some data is personal and owned by them, and the importance of keeping their personal data safe. Children explore steps to take when encountering inappropriate content, pop-ups or uninitiated contact online. Children follow an algorithm (sequence of steps) to achieve an outcome, and they begin to understand computational thinking through sequence.

In Digital Technologies, children explore needs for design, and develop, create and evaluate designed solutions.

Content Descriptions

Digital systems

Digital systems have common features, including hardware devices and software, and are used at home, in school and in the community

For example:

- digital systems can include mobile devices, tablets and desktop personal computers (PCs)
- different digital systems are used to capture or transfer data (taking a photograph, voice or video)
- common features of digital systems include screens, camera, speakers and buttons that are usually integrated
- digital technologies are frequently networked or connected, enabling people to share, communicate, or store data

WAPDIGDS1

Digital literacy

Data representation

Data can be represented as objects and images

For example:

- icons are images that represent programs or applications (apps)
- street signs are images that represent data
- objects and events can be sorted based on easily identified characteristics and can be represented using digital systems, such as birthdates and categories
- a sun picture could represent 'hot' weather, an arrow image could represent movement in a particular direction

WAPDIGDR1

Digital literacy

Privacy and security

Some data is personal and owned by them

For example:

- some data is personal, such as photographs of themselves with their families
- some data is public, such as photographs of local community sites
- photographs of students may become public, such as their username in a game or photographs of them on a parent's social media account
- ask a friend for permission before taking their photograph
- personal data can be obtained and used by nefarious strangers

WAPDIGPS1

Ethical understanding Digital literacy

Steps to take when encountering inappropriate content, pop-ups, or uninitiated contact

WAPDIGPS2

Ethical understanding Digital literacy

Digital implementation

Follow an algorithm (sequence of steps) to achieve an outcome

WAPDIGDI1

Digital literacy

Design thinking skills

Project management

Share ideas to develop a solution

WAPDIGDTPM1

Personal and social capability

Investigating and defining

Explore the purpose for design

WAPDIGDTID1

Critical and creative thinking

Designing

Design solutions through discussion, drawing and/or modelling to meet a personal need

WAPDIGDTDE1

Critical and creative thinking

Producing and implementing

Use available technologies and materials to safely create a solution

WAPDIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Use personal preferences to evaluate the solution

WAPDIGDTEV1

Literacy Critical and creative thinking

Year 1

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning experiences are informed by the Principles and Practices of the Early Years Learning Framework. A holistic curriculum that integrates knowledge, understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Digital Technologies provides opportunities for children to explore digital systems and digital technologies use, while developing an understanding of what the internet is, participating in digital contexts safely and developing foundational skills in computational thinking.

In Year 1, children have opportunities to create a range of solutions through guided learning, focusing on the use of digital systems comprised of hardware and software that are used together to achieve a common goal. Children explore ideas and design opportunities, and identify important information from data. They learn to follow a visual representation of a sequence of steps through flowcharting.

Children learn that data can be represented as images, symbols, numbers and words, and that some data is personal, owned by them and should be shared only with trusted people. They access their school account, with assistance. Children use available technologies to safely manage and create a preferred solution, and evaluate these solutions based on personal preferences.

Achievement Standard

By the end of the year:

Children identify specific features of digital systems (hardware and software) and how they are used in everyday life. They

represent data using pictures, symbols, numbers and words. Children follow strategies to stay safe online, while identifying that data is personal and can be shared with trusted people. They access their school account, with assistance, using a recorded username and password.

In Digital Technologies, children explore ideas and design opportunities when designing products or solutions. They follow a visual representation of an algorithm. Children design solutions through drawing, modelling and/or a sequence of steps to plan and produce solutions for a personal need. They use personal preferences to evaluate the solution for a personal need.

Content Descriptions

Digital systems

Digital systems have hardware and software that are used together

For example:

- digital systems include mobile devices, tablets and desktop personal computers (PCs)
- digital systems are a collection of hardware and software
- hardware is physical and software is non-physical
- software includes applications, games, operating systems
- digital systems can be used for downloading and storing information, or for a purpose, such as retelling a story

WA1DIGDS1

Digital literacy

Data representation

Data can be represented as images, symbols, numbers and words

For example:

- the equivalence of different representations of numbers, such as words, digits, numbers and dice dots
- representing categorical data in a variety of ways, such as drawings, lists and tally marks
- the relationship between simple representations, such as arrows or appropriate emoji, and the concept or emotion they represent
- cultural symbols can represent people, objects and movement

WA1DIGDR1

Numeracy Digital literacy

Privacy and security

Some data is personal, owned by them and can be shared with trusted people

For example:

- importance of asking permission from a parent or carer before entering personal details online, such as address, phone number and date of birth
- websites and apps used at home and school collect personal data, such as usernames and email addresses
- ask a friend before sharing a photograph of them

WA1DIGPS1

Ethical understanding Digital literacy

Access their school account, with assistance, using a recorded username and password

WA1DIGPS2

Digital literacy

Digital implementation

Follow a visual representation of an algorithm (sequence of steps)

WA1DIGDI1

Digital literacy

Design thinking skills

Project management

Share ideas and work with others to develop a solution

WA1DIGDTPM1

Literacy Personal and social capability

Investigating and defining

Explore ideas and design opportunities for a personal need

WA1DIGDTID1

Critical and creative thinking

Designing

Design solutions through drawing, modelling and/or a sequence of steps

WA1DIGDTDE1

Critical and creative thinking

Producing and implementing

Use available technologies and materials to safely create a solution

WA1DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Use personal preferences to evaluate the solution for a personal need

WA1DIGDTEV1

Literacy Critical and creative thinking

Year 2

Year Level Description

In the early childhood phase of schooling, learning, development and wellbeing are connected and learning experiences are informed by the Principles and Practices of the Early Years Learning Framework. A holistic curriculum that integrates knowledge, understandings, skills, values and attitudes across learning areas connects learning to children's lives and their natural curiosity about their world.

Digital Technologies provides opportunities for children to explore digital systems and digital technologies use, while developing an understanding of what the internet is, participating in digital contexts safely and developing foundational skills in computational thinking.

In Year 2, children are provided opportunities to apply computational thinking by creating and following algorithms that include a sequence of instructions and decisions, and by using digital systems to produce solutions. Through investigation and practice, they develop confidence in representing data in different ways. Children discover how information systems meet information, communication and/or recreational needs. They explore the concept that personal data may be safely shared online with specific people using trusted platforms, while practising how to safely access their school account.

Children develop their design skills by conceptualising algorithms as a sequence of steps for carrying out instructions, such as identifying steps in a process or controlling robotic devices. They have the opportunity to create a range of solutions through guided learning and in collaboration with peers.

Achievement Standard

By the end of the year:

Children use digital systems for a specific purpose, making connections between software and hardware. They identify patterns within data and recognise that data can be represented by diagrams, symbols, numbers and words. Children independently access their school account with a recorded username and password and log out afterwards. Children recognise that websites and apps store personal data, and identify behaviours to stay safe online.

In Digital Technologies, children explore design opportunities to meet needs. They develop, follow and describe algorithms and decisions made by the user. They use given equipment to safely produce solutions. Children evaluate the success of design ideas and solutions based on the needs of known users.

Content Descriptions

Digital systems

Digital systems, including hardware devices and software, are used for an identified purpose

For example:

- digital systems, such as mobile devices, tablets and desktop personal computers (PCs) are used/selected for a purpose
- digital systems are a collection of hardware and software
- hardware is physical and software is non-physical
- software includes applications, games, operating systems
- different software applications have different uses

WA2DIGDS1

Digital literacy

Data representation

Data can have patterns and may be represented as diagrams, symbols, numbers and words

For example:

- data can be represented using a variety of visualisation techniques, such as picture graphs
- data patterns can be exemplified by the repetition of pictures and symbols
- number patterns can be represented as pairs or in multiples of two

WA2DIGDR1

Numeracy Digital literacy

Privacy and security

Some personal data may be safely shared online with specific people using trusted platforms

For example:

- using a school learning app to share photographs or videos with a parent
- personal data, such as usernames or avatars in online games, is visible on some websites and apps

WA2DIGPS1

Ethical understanding Digital literacy

Independently access their school account with a recorded username and password, and log out

WA2DIGPS2

Digital literacy

Digital implementation

Create an algorithm (sequence of steps) including decisions made by the user

WA2DIGDI1

Digital literacy

Follow algorithms (sequence of steps) including decisions made by the user

WA2DIGDI2

Digital literacy

Design thinking skills

Project management

Plan, share ideas and work with others to develop a solution for a known user

WA2DIGDTPM1

Literacy Personal and social capability

Investigating and defining

Explore ideas and design opportunities for a known user

WA2DIGDTID1

Critical and creative thinking

Designing

Design solutions through discussion, drawing, modelling and/or a sequence of steps

WA2DIGDTDE1

Critical and creative thinking

Producing and implementing

Use given equipment and technologies to safely create a preferred solution

WA2DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Use personal preferences and the needs of the known user to evaluate the solution

WA2DIGDTEV1

Literacy Critical and creative thinking

Year 3

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Digital Technologies builds on concepts previously acquired, and students continue to develop understanding and skills in computational thinking.

In Year 3, students explore digital systems in terms of their components and peripheral devices. They develop an understanding of the characteristics of data and its representation. Students have opportunities to create digital solutions, such as interactive adventures and simple guessing games that involve user choice.

Students apply design thinking skills to generate multiple ideas for their designed solutions. They reflect on their learning and work practices and consider ways in which these might be improved, modified or adapted for different situations. Students develop their design skills by following prepared algorithms that include a choice of options through branching. They experiment with appropriate software, including visual programming environments to implement their solutions.

Achievement Standard

By the end of the year:

Students explore and recognise peripheral devices and their purpose, identify that there are different types of data and that data can be represented, stored and shared online in different ways. They access their school account, using a unique, private, memorised password, and log out afterwards.

In Digital Technologies, students create, design and implement algorithms in a visual programming environment to include decisions made by the user to solve a given digital task. They develop and communicate ideas using labelled drawings and technical terms. Students select and safely use appropriate components with given equipment to create a solution. They design and communicate solutions, and follow a plan to produce designed solutions. Students use criteria to evaluate developed design processes and solutions.

Content Descriptions

Digital systems

Digital systems and peripheral devices are connected and used together for various purposes

For example:

- input peripheral devices include keyboard, mouse, camera and microphone
- output peripheral devices include monitor, printer, 3D printer, speaker and remotes
- peripherals that have the ability to be both input and output devices include touch screen, headsets and controllers

WA3DIGDS1

Digital literacy

Data representation

Data is of different types and can be represented in various ways

For example:

- data is of different types including sound, images, numbers and text
- symbols and icons are used to represent data
- symbolic representations such as flowcharts
- infographics that combine images, symbols, and diagrams tell a compelling data story and provide information

WA3DIGDR1

Numeracy Digital literacy

Privacy and security

Different types of personal data are shared and stored online

For example:

- personal data stored in accounts at school and at home
- consider who has access to data, such as documents in school cloud storage that are accessible by the teacher, or a nickname in their online gaming accounts is visible to all players

WA3DIGPS1

Ethical understanding Digital literacy

Access their school account using a unique, private, memorised password, and log out afterwards

WA3DIGPS2

Digital literacy

Digital implementation

Represent algorithms (sequence of steps), including decisions made by the user (branching) using flow charts

WA3DIGDI1

Digital literacy

Implement algorithms (sequence of steps) in a visual programming environment to include decisions made by the user (branching)

WA3DIGDI2

Numeracy Digital literacy

Design thinking skills

Project management

Communicate ideas and follow a plan, with consideration of time management, to develop a solution

WA3DIGDTPM1

Literacy Personal and social capability

Investigating and defining

Define ideas and design opportunities for individual and/or local needs

WA3DIGDTID1

Critical and creative thinking

Designing

Design solutions through use of labelled drawings, technical terms and/or a sequence of steps

WA3DIGDTDE1

Critical and creative thinking

Producing and implementing

Use appropriate technologies and components with given equipment and follow agreed protocols to produce a designed solution

WA3DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Use given criteria to evaluate diagrams, technologies and the components used for the designed solution

WA3DIGDTEV1

Literacy Critical and creative thinking

Year 4

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Digital Technologies builds on concepts previously acquired, and students continue to develop understanding and skills in computational thinking, such as categorising and outlining procedures. They have opportunities to create a range of solutions, such as interactive adventures that involve user choice, modelling simplified real-world systems.

In Year 4, students explore digital systems in terms of peripheral devices. They experiment with refining their designing skills, representing algorithms that support decisions (branching) and repetition using flow charts. Students explain the risks of personal data that is shared and stored online. They experiment with digital systems to investigate the advantages of different representational forms and different technologies, their purposes and situations.

Students implement solutions using appropriate software, including visual programming environments that use a variety of graphical elements. They design solutions to meet specific needs and consider how society uses digital systems to meet community requirements.

Achievement Standard

By the end of the year:

Students identify different purposes for digital systems, hardware components and peripheral devices. They identify how data can be represented in a range of ways. In a visual programming environment, students design and implement algorithms that involve decisions and repetition, and create and communicate ideas and information. Students identify that personal data that is shared and stored online can pose risks. They access their school account, using a memorised password, and describe the risks of not logging out.

In Digital Technologies, students develop and communicate design ideas and decisions, using labelled drawings and technical terms. Students use agreed protocols and management roles to plan, make decisions and communicate ideas to develop solutions. They use given criteria to evaluate design features, selected resources, their decision-making process and the designed solution for a given digital task.

Content Descriptions

Digital systems

Digital systems, including peripheral devices, are used to transfer and store different types of data

For example:

- peripheral devices, such as a keyboard, touch screen, mouse, camera, microphone, monitor, printer, 3D printer, speaker, storage, gaming controller and headset can be categorised as input, output and storage functions
- peripheral devices can be wired or wirelessly connected

WA4DIGDS1

Digital literacy

Data representation

Data of the same type can be represented in different ways depending on the purpose

For example:

- data can be of the same type including sound, images, numbers and text
- different types of data can be used depending on the purpose or needs, such as numbers, letters, symbols, pictures or sounds
- identifying circumstances when the same data can be represented in different ways and why some representations are better than others in certain contexts, such as four vs 4 vs IV vs |||| vs quatre, and that numbers are preferred for calculation than words

WA4DIGDR1

Numeracy Digital literacy

Privacy and security

Personal data that is shared and stored online can pose risks

For example:

- personal data stored in online accounts forms a person's digital identity and reveals detailed information about them, such as photographs which can reveal details about their location, habits or home
- personal data, when shared online, cannot be removed
- age restrictions identified in terms and conditions of websites and apps

WA4DIGPS1

Ethical understanding Digital literacy

Access their school account, using a memorised password. It should be easy to remember but difficult for others to guess. Risks of not logging out

WA4DIGPS2

Digital literacy

Digital implementation

Represent an algorithm (sequence of steps) involving decisions (branching) and repetition using flow charts

WA4DIGDI1

Numeracy Digital literacy

Implement algorithms (sequence of steps) in a visual programming environment to include decisions (branching) and repetition

WA4DIGDI2

Numeracy Digital literacy

Design thinking skills

Project management

Use agreed protocols and management roles to communicate ideas, plan and make decisions to develop solutions

WA4DIGDTPM1

Literacy Personal and social capability

Investigating and defining

Define the features of a design brief and the requirements of a design task for a community need

WA4DIGDTID1

Critical and creative thinking

Investigate and select resources based on properties for the given task

WA4DIGDTID2

Critical and creative thinking

Designing

Design solutions through use of labelled drawings, technical terms, decision-making and/or a sequence of steps

WA4DIGDTDE1

Critical and creative thinking

Producing and implementing

Use appropriate technologies, components and/or equipment and follow agreed protocols to produce a designed solution

WA4DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Use given criteria to evaluate design features, selected resources, decision-making processes and the designed solution

WA4DIGDTEV1

Literacy Critical and creative thinking

Year 5

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Digital Technologies builds on concepts previously acquired where students continue to develop understanding and skills in computational and design thinking.

In Year 5, students explore the role that individual components of digital systems play in the processing and representation of data. They recognise that all digital systems represent data as numbers. Students create a range of digital solutions that involve user input, variables and control structures.

When creating solutions, students identify appropriate data and requirements. They evaluate how well their design solutions work in current digital systems and reflect on their design process. Students develop strategies to communicate information and ideas using agreed protocols, considering the safety aspects of working in digital environments while identifying websites and apps that are trusted to store personal data online. They demonstrate increased responsibility for managing and organising activities, individually and in groups.

Achievement Standard

By the end of the year:

Students identify internal components of digital systems and their functions. They identify ways data is represented using code, and provide relevant examples. Students create designed solutions, follow and represent diagrammatic algorithms involving user input, variables and control structures, such as sequencing, decisions and repetition. They implement algorithms using visual programming languages. Students identify and make judgements based on the use of personal data and how it contributes to a permanent digital footprint. They access multiple personal accounts using unique passphrases or biometrics and describe the risks of password reuse and not logging out.

In Digital Technologies, students define a problem, identify available resources and develop designed solutions. They develop and communicate alternative solutions, and use annotated diagrams, storyboards and appropriate technical terms for design ideas.

Content Descriptions

Digital systems

Digital systems have main internal components that perform particular functions to achieve a purpose

For example:

- common internal hardware components, such as a central processing unit (CPU), random access memory (RAM), motherboard, solid state drive (SDD) and hard disk drive (HDD), are interconnected and work together to form a digital system
- data is transmitted between digital systems in different ways such as wires, cables, radio waves

WA5DIGDS1

Digital literacy

Data representation

Data of all types, including text, numbers, sound and images, are represented using codes

For example:

- data can be represented using whole numbers in a digital system, such as converting letters in a message to numbers using position in the alphabet
- the encoding and transmitting of data over a distance, such as Morse Code, semaphore, fire signals, drumming and radio
- digital systems represent data, such as Unicode or ASCII

WA5DIGDR1

Numeracy Digital literacy

Privacy and security

Personal data can be used to create a permanent digital footprint

For example:

- individuals leave a digital footprint which includes their activity on social media, online searches and communication platforms
- the ability of websites and apps to safely store data and the level of trustworthiness can vary
- importance of protecting someone's privacy and only collecting data when required, such as choosing not to collect information about someone's birthdate when it is not necessary reduces the risk of private data being stolen in a cyber attack

WA5DIGPS1

Ethical understanding Digital literacy

Access multiple personal accounts using unique passphrases or biometrics. Risks of password reuse and not logging out

WA5DIGPS2

Digital literacy

Digital implementation

Design algorithms in plain English and/or flow charts that involve user input, variables and control structures (sequence, decisions and repetition)

WA5DIGDI1

Numeracy Digital literacy

Implement algorithms in a visual programming environment involving variables and control structures (sequence, decisions and repetition) with user input

WA5DIGDI2

Numeracy Digital literacy

Design thinking skills

Project management

Use agreed protocols and management roles to communicate decisions, plan and manage time to develop designed solutions

WA5DIGDTPM1

Literacy Personal and social capability

Investigating and defining

Break down a design brief to define the purpose and requirements for a given task

WA5DIGDTID1

Critical and creative thinking

Investigate and select resources based on properties and functions for the given task

WA5DIGDTID2

Critical and creative thinking

Designing

Design solutions considering competing factors, with annotated diagrams, storyboards and/or a sequence of steps, using technical terms and an iterative process

WA5DIGDTDE1

Critical and creative thinking

Producing and implementing

Use technologies, components and/or equipment to implement agreed protocols to produce a designed solution

WA5DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Use given criteria to evaluate design features, with consideration of competing factors, processes and the designed solution

WA5DIGDTEV1

Literacy Critical and creative thinking

Year 6

Year Level Description

In the middle to late childhood phase of schooling, students develop a sense of self, their world expands, and they begin to see themselves as members of larger communities. Learning experiences emphasise and lead to an appreciation of both the commonality and diversity of human experience and concerns.

Digital Technologies builds on concepts previously acquired where students continue to develop understanding and skills in computational and design thinking.

In Year 6, students consolidate their understanding of the role individual components of digital systems play in the processing and representation of data. They are introduced to wired and wireless networks of digital systems that send data in binary form. Students are introduced to the concept of digital footprints and consider privacy considerations when collecting user data.

Students further develop abstractions by identifying common elements across similar problems and systems and make connections between models and the real-world systems they represent. They are given increased responsibility for managing and organising activities, individually and in groups. Students apply systems thinking when investigating the functions and purpose of each component in a digital system and in their interactions with others. When creating digital solutions, students further refine their skills to identify and use appropriate data and requirements. They incorporate control structures when

implementing their solutions through visual programming environments.

Achievement Standard

By the end of the year:

Students outline interactions between wired and wireless networks to transmit data for a variety of purposes. They explain how data can be represented by off and on states (zeros and ones in binary) and make simple conversions. Students use visual programming environments to design, modify and implement algorithms that involve user input, variables and control structures such as sequence, decisions and various types of iteration. They identify their digital footprint and privacy considerations when collecting user data. Students access multiple personal accounts using unique passphrases or biometrics. They describe the risks of password reuse and practices that reduce risk to their personal accounts.

In Digital Technologies, students identify available resources to design a solution for a given digital task, using critical thinking strategies to make decisions. They develop, design and compare alternative solutions, achieved through an iterative process including graphical representations, use of a range of technologies, techniques, technical terms and/or a sequence of steps. They use agreed protocols to set goals, manage competing factors, resources and time, to plan, develop and communicate ideas and solutions.

Content Descriptions

Digital systems

Digital systems are connected in wired and wireless networks to transmit data for a variety of purposes

For example:

- familiar networks can be found in the school, home or local community and can be connected through the internet
- wired and wireless networks have differences, such as speed, device mobility and ease of installation
- separate systems can be connected in different ways to exchange data
- data is transmitted through a network, broken up into packets (small pieces) and passed from the source, through multiple devices, to the destination

WA6DIGDS1

Digital literacy

Data representation

Data can be represented by on and off states (zeros and ones in binary)

For example:

- on and off states in a circuit can represent the digits one and zero, and is how digital systems represent data
- converting binary to decimal and vice versa
- images are represented in digital systems using binary

WA6DIGDR1

Numeracy Digital literacy

Privacy and security

Digital footprint and privacy considerations when collecting user data

For example:

- individuals leave a digital footprint which includes their activity on social media, online searches and communication platforms
- data and images that have been posted online can lead to information resurfacing at a later date, such as a comment made on a social media post or video can associate a person with their comment and the content
- when providing data to websites it is important to know how that data may be stored or used in the future and if this poses a risk
- sharing personal information increases the likelihood it will be revealed in the future

WA6DIGPS1

Ethical understanding Digital literacy

Access multiple personal accounts using unique passphrases or biometrics. Risks of password reuse and practices to reduce risk to their personal accounts

WA6DIGPS2

Digital literacy

Digital implementation

Design algorithms in plain English and/or flow charts that involve user input, variables and control structures (sequence, decisions and various types of iteration: For, Repeat, While)

WA6DIGDI1

Numeracy Digital literacy

WA6DIGDI2 Implement algorithms in a visual programming environment involving variables and control structures (sequence, decisions, input and various types of iteration)

Numeracy Digital literacy

Design thinking skills

Project management

Use agreed protocols to set goals, manage competing factors, resources and time, to plan, develop and communicate decisions when developing designed solutions

WA6DIGDTPM1

Literacy Personal and social capability

Investigating and defining

Break down a design brief to define the purpose, requirements and constraints for a given task

WA6DIGDTID1

Critical and creative thinking

Investigate and select resources considering constraints, properties and functions appropriate for the given task

WA6DIGDTID2

Critical and creative thinking

Designing

Design alternative solutions achieved through an iterative process, including critical thinking, graphical representations, use of a range of technologies, techniques, technical terms and/or a sequence of steps

WA6DIGDTDE1

Critical and creative thinking

Producing and implementing

Use a range of technologies, components and/or equipment to implement agreed protocols to produce a designed solution

WA6DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Develop negotiated criteria to evaluate design features, graphics, selected technologies, processes and functionality, with consideration of constraints for the designed solution

WA6DIGDTEV1

Literacy Critical and creative thinking

Year 7

Year Level Description

In the early adolescence phase of schooling, students align with their peer group and begin to question established conventions, practices and values. Learning and teaching programs assist students to develop a broader and more comprehensive understanding of the contexts of their lives and the world in which they live.

Digital Technologies further develops student understanding and skills in computational and design thinking, such as decomposing problems, and engages students with a wider range of information systems. Students begin to develop an interest in particular fields of knowledge.

In Year 7, students create a range of digital solutions. They explore the properties and hardware devices of networked systems. Students acquire, store and visualise data from a range of sources using spreadsheets. They further develop their understanding of the vital role that data plays in their lives.

When defining problems, students identify the key elements, factors and constraints at play. They design and develop increasingly complex algorithms. Students predict and evaluate their developed and existing solutions, considering time, tasks, data and the safe use of information systems. They plan and manage individual and team projects with some autonomy.

Students consider ways of managing the exchange of ideas, tasks, files and feedback.

Achievement Standard

By the end of the year:

Students identify the ways different types of networks, including wired, wireless and mobile networks, and their hardware components, transmit data. They identify how digital systems represent data using binary. Students acquire, store and visualise data from a range of sources using spreadsheets. They design algorithms involving control structures (iteration and selection). Students represent these algorithms using flow charts, pseudocode and correct terminology, and use them to implement, modify and debug programs. Students identify issues relating to a user's digital footprint and the permanence of data, while outlining ways of protecting accounts with multifactor authentication.

In Digital Technologies, students use project management processes and skills to plan, develop and communicate solutions, while considering time and available resources, and identify the purpose for a given digital task. Students use a range of techniques, technologies and appropriate terminology to design, develop, review and communicate ideas, plans and processes. They implement agreed protocols when using a range of technologies, components and/or equipment to produce designed solutions.

Content Descriptions

Digital systems

Methods of data transmission in different types of networks including wired, wireless and mobile networks

For example:

- wireless and wired connectivity has advantages and disadvantages
- mobile networks transfer data, and transmission rates differ between mobile, wired and wireless (wi-fi) networks
- benefits of cloud storage include scalability, accessibility, security and cost-effectiveness

WA7DIGDS1

Digital literacy

Hardware devices of networks and their purposes

For example:

- common hardware devices in networks include router (home and corporate), switch and server
- networks including their devices and transmission media can be illustrated through the use of simple network diagrams
- IoT (Internet of Things) devices include sensors, actuators, etc. to connect and exchange data typically wirelessly with other devices, such as fridges, to increase usability and functionality

WA7DIGDS2

Digital literacy

Data representation

Digital systems use binary to represent data in text

For example:

- binary is used to represent electrical signals inside a computer circuit or system
- whole numbers can be represented in binary, recognising one byte = 8 bits, which can represent values from 0 to 255
- digital systems represent text as a sequence of individual characters numbered using the Unicode character set, such as upper-case and lower-case letters, punctuation and emojis
- digital systems represent data in binary, such as by converting a character to its Unicode or ASCII value, then converting that value into binary

WA7DIGDR1

Numeracy Digital literacy

Acquiring, managing and analysing data

Acquire, store and visualise data from a range of sources using spreadsheets

For example:

- datasets can be visualised by choosing appropriate graphs using spreadsheets, such as a scatter plot of food prices and sales coloured by each food's sugar content, or diagrams, like a social network diagram, or mapping census data by location

WA7DIGAD1

Numeracy Digital literacy

Privacy and security

Issues relating to a user's digital footprint and the permanence of data

For example:

- data and/or images posted online can lead to information resurfacing at a later date, such as a social media post or video associating a person with both their comment and the content
- a Google search can reveal what data is stored about oneself through searching by name, specific details, email address or social media profile names

WA7DIGPS1

Ethical understanding Digital literacy

Protecting accounts with multifactor authentication

WA7DIGPS2

Literacy Digital literacy

Digital implementation

Break down the user experience (UX) of a digital system

WA7DIGDI1

Critical and creative thinking Digital literacy

Design algorithms involving control structures (selection, decision and iteration), and represent them using flow charts and pseudocode

WA7DIGDI2

Numeracy Digital literacy

Implement, modify, and debug programs involving control structures

WA7DIGDI3

Digital literacy

Design thinking skills

Project management

Plan, develop and communicate, using project management processes, considering time and available resources to achieve solutions

WA7DIGDTPM1

Personal and social capability

Investigating and defining

Investigate and define the problem and requirements of a given design brief

WA7DIGDTID1

Literacy Critical and creative thinking

Break down a given design brief, identifying and defining the purpose and competing considerations

WA7DIGDTID2

Critical and creative thinking

Consider given technologies, resources and/or components to develop solutions

WA7DIGDTID3

Critical and creative thinking

Designing

Design processes and solutions with given technologies and techniques, using appropriate technical terms

WA7DIGDTDE1

Critical and creative thinking

Producing and implementing

Implement agreed protocols and use a range of technologies, components and/or equipment to produce designed solutions

WA7DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Use given contextual criteria to evaluate design processes and solutions

WA7DIGDTEV1

Literacy Critical and creative thinking

Year 8

Year Level Description

In the early adolescence phase of schooling, students align with their peer group and begin to question established conventions, practices and values. Learning and teaching programs assist students to develop a broader and more comprehensive understanding of the contexts of their lives and the world in which they live.

Digital Technologies further develops student understanding and skills in computational and design thinking, such as decomposing problems, and engages with a wider range of information systems as they broaden their experiences and involvement in local, regional, national and global activities.

In Year 8, students investigate the properties of networked systems and their suitability, based on performance specifications. They consider methods of data transmission and security in wired, wireless and mobile networks. They analyse and validate data using spreadsheets and make predictions by identifying trends. Students continue to investigate patterns, processes and phenomena, and explore forms of data representation and technology.

When defining problems, students identify the key elements of the problems and the factors and constraints at play. They design increasingly complex algorithms. Students develop an understanding of ethical issues regarding the collection and ownership of data.

Achievement Standard

By the end of the year:

Students describe methods of data transmission and security in wired, wireless and mobile networks, identify specifications of hardware components and outline potential impact on particular tasks. They identify how digital systems represent image and audio data using binary. Students analyse and validate data, using spreadsheets, to draw conclusions and make predictions based on identified trends. They design algorithms involving nested control structures, represented by flow charts and pseudocode, in plain English. Students implement, modify and debug programs involving control structures in a general-purpose programming language. They trace algorithms to predict output for a given input and identify errors. Students identify ethical issues regarding the collection and ownership of data and evaluate the authenticity, accuracy and timeliness of acquired data. They describe cybersecurity threats, including phishing.

In Digital Technologies, students investigate a given need or opportunity for a specific purpose and develop a design brief. Students consider and select components and resources to develop solutions, identifying constraints. They use appropriate technical terms and technologies to design, develop, evaluate and communicate alternative digital solutions. Students use project management skills to plan, develop and communicate, while considering time, resources and costs to achieve solutions. They develop contextual criteria to evaluate design processes and solutions.

Content Descriptions

Digital systems

Methods of data transmission and security in wired, wireless and mobile networks

For example:

- data is structured and transmitted through a network via data packets (small pieces), and passed from the source through multiple devices, in order, to the destination
- strategies are necessary to mitigate common security threats, such as phishing
- common security strategies including VPNs, antivirus software, encryption, multifactor authentication, etc.

WA8DIGDS1

Digital literacy

The effect of hardware specifications on performance and the appropriateness of hardware for particular tasks

For example:

- appropriate hardware is selected for particular tasks, such as choosing a powerful graphics card for computer gaming or large external storage for video editing
- network properties, such as the bandwidth, latency and reliability of wired, wireless and mobile networks can be compared

WA8DIGDS2

Digital literacy

Data representation

Digital systems represent image and audio data using binary

For example:

- digital systems represent bitmap images, such as PNG and JPEG, as the colour of each pixel in separate red, green and blue (RGB) channels ranging from 0 to 255, and represent vector graphics, such as scalable vector graphics (SVG) using the geometry of lines and shapes
- digital systems represent audio using whole numbers for the amplitude of soundwaves

WA8DIGDR1

Numeracy Digital literacy

Acquiring, managing and analysing data

Analyse and validate data using spreadsheets to draw conclusions and make predictions by identifying trends

For example:

- data based on its attributes can be used to identify trends and make predictions
- spreadsheets are used to filter and sort sports data in order to predict future trends
- spreadsheet functions such as MIN, MAX, COUNT, COUNTIF, MEAN, MODE, MEDIAN can be used to analyse data
- critical thinking and scepticism should be employed when encountering data from unfamiliar or dubious sources such as 'deepfakes'

WA8DIGAD1

Numeracy Digital literacy

Evaluate the authenticity, accuracy and timeliness of acquired data

For example:

- critical thinking and scepticism should be employed when encountering data, such as ‘deepfakes’, from unfamiliar or dubious sources

WA8DIGAD2

Numeracy Digital literacy

Privacy and security

Ethical issues relating to the collection and ownership of data

For example:

- the importance of verifying the source and origin of data, in conducting research to assess the credibility and trustworthiness of data providers. Encourage critical thinking and scepticism when encountering data from unfamiliar or dubious sources
- fact-checking methodologies and tools to cross-reference sources and evaluate the authenticity of data collected

WA8DIGPS1

Ethical understanding Digital literacy

Cybersecurity threats including phishing

WA8DIGPS2

Numeracy Digital literacy

Digital implementation

Design the user experience (UX) of a digital system

WA8DIGDI1

Critical and creative thinking Digital literacy

Design algorithms involving nested control structures and represent them using flow charts and pseudocode

WA8DIGDI2

Numeracy Digital literacy

Trace algorithms to predict output for a given input and to identify and fix errors

WA8DIGDI3

Numeracy Digital literacy

Implement, modify and debug programs involving control structures in a general-purpose programming language

WA8DIGDI4

Digital literacy

Design thinking skills

Project management

Plan, develop and communicate, using project management processes, considering time, resources and costs to achieve solutions

WA8DIGDTPM1

Personal and social capability

Investigating and defining

Investigate a problem for a given need or opportunity

WA8DIGDTID1

Literacy Critical and creative thinking

Develop a design brief for a given need or opportunity

WA8DIGDTID2

Critical and creative thinking

Consider technologies, resources and/or components to develop solutions, identifying constraints

WA8DIGDTID3

Critical and creative thinking

Designing

Design processes and solutions considering a range of technologies and techniques, using appropriate technical terms

WA8DIGDTDE1

Critical and creative thinking

Producing and implementing

Implement agreed protocols, a range of technologies, techniques, components and processes to produce designed solutions

WA8DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Use student-developed contextual criteria to evaluate design processes and solutions

WA8DIGDTEV1

Literacy Critical and creative thinking

Year 9

Year Level Description

In the middle adolescence phase of schooling, teaching and learning programs encourage students to develop an open and questioning view of themselves as active participants in their society and the world.

Digital Technologies further develops student understanding and skills in computational thinking, such as the precise and accurate describing of problems, and the use of functions to create solutions. It also focuses on engaging students with specialised learning to further develop their critical and design thinking skills.

In Year 9, students consider the role of hardware and software in securing the movement of data in a digital system. Students explore different methods of manipulation, storage and transmission of data, and students develop abstractions, identifying standard elements and modelling entities and their relationships. They interrogate security practices and techniques used to compress data and learn about the importance of separating content, presentation and behavioural elements for data integrity and maintenance purposes.

When defining problems, students consider the functional and non-functional requirements of a solution through interacting with users and reviewing processes. They consolidate their algorithmic design skills to incorporate testing. Students develop solutions to problems and evaluate their solutions and existing information systems, based on a set of criteria. They consider the privacy and security implications of how data is used and controlled, and suggest how policies and practices can be improved to ensure the sustainability and safety of information systems.

Achievement Standard

By the end of the year:

Students describe the role of hardware and software to secure the movement of data in digital systems with a focus on cybersecurity threat models. They identify different methods used for manipulation, storage and transmission of data. Students acquire, store and validate data from a range of sources using software, including spreadsheets and databases. They define and

decompose real-world problems by interviewing stakeholders to develop user experiences. Students design and prototype the user experience of a digital system. They design algorithms that use functions and represent them as flow charts and/or pseudocode. Students implement and modify programs in a general-purpose programming language and trace algorithms to predict output against a range of test cases. They describe the Australian Privacy Principles regarding the collection and ownership of data.

In Digital Technologies, students identify and define the needs of a stakeholder to develop a design brief for a solution. They investigate a range of components and resources to develop ideas and identify and consider constraints. Students apply design thinking, creativity and enterprise skills to produce purposeful and holistic solutions. Students select, test and implement appropriate technologies and processes to produce effective solutions and evaluate design processes against student-developed criteria. Students manage projects using digital technologies through an agile and collaborative approach, and consider time, risk, economic and sustainable factors.

Content Descriptions

Digital systems

Role of hardware and software to manage, control and secure the movement of data in a digital system

For example:

- simple network configurations using real or simulated hardware allows for the observation of packets moving around the network, such as monitoring packets on simulated switches and networked devices
- domain names and IP addresses allow data to be transmitted to specific networked devices, such as DNS and routing tables
- the IoT (Internet of Things) is used as part of a networked digital system, such as using sensors and digital systems to collect and share data over the internet with identified security risks
- common network security mitigation strategies include firewalls, intrusion detection software, etc
- ping and/or traceroute commands show network connectivity

WA9DIGDS1

Digital literacy

Data representation

Different methods of manipulation and storage of data

For example:

- video and sound manipulation software can affect file storage
- different image manipulation techniques (compression, cropping, exporting, etc.) have effects on file sizes
- file sizes are larger for augmented reality (AR) and virtual reality (VR) and can affect the amount of storage required
- manipulation of images/videos can occur through the use of artificial intelligence, such as 'deepfakes'

WA9DIGDR1

Digital literacy

Data compression techniques for an intended purpose

For example:

- algorithms can be used to identify patterns in data and represent them in a compressed way, such as repeated pixels in an image with run length encoding
- explore the difference between 'lossy and lossless' compression and the consequences of each, by exploring audiovisual compression and the impact of different formats, such as MP3, MP4, JPEG, WAV or RAW on file size and quality
- virtual reality (VR) file types include FBX, OBJ, GLTF and VRML/X3D

WA9DIGDR2

Numeracy Digital literacy

Acquiring, managing and analysing data

Acquire, store and validate data from a range of sources using software, including spreadsheets and/or databases

For example:

- different methods of data collection such as surveys, face-to-face interviews, phone interviews, observation, comments in response to a social media posting, phone logs, browser history and online webcam systems have strengths and weaknesses
- accessing, storing and manipulating data from the Australian Bureau of Statistics in a format that is useful for analysis, such as using a spreadsheet to acquire, filter, group and sort data on population growth across age groups in Australia
- systems can be developed that check data is correct and meaningful using automated techniques and manual analysis, such as validating movie review data using validation rules and input forms
- ensuring that the data used to train an AI model is representative and accurate to minimise any potential biases of the output

WA9DIGAD1

Numeracy Digital literacy

Single table (flat file) databases are created to store and manage data

For example:

- single table (flat file) databases are used to store data in a structured manner with tables, records, fields and primary keys

WA9DIGAD2

Numeracy Digital literacy

Privacy and security

Australian Privacy Principles (APP) regarding the collection and ownership of data

For example:

- online services allow control access to user data in line with the APPs, such as assessing whether users' social media accounts allow for them to update their contact information, and who can see that information on the platform
- APPs can be used to reference and evaluate the steps users take to protect information, such as how companies store user data to prevent security breaches
- websites contain individual privacy policies that differ and can be investigated

WA9DIGPS1

Ethical understanding Digital literacy

Cybersecurity threat models

WA9DIGPS2

Numeracy Digital literacy

Digital implementation

Define and decompose real-world problems by surveying stakeholders to create the requirements of the user

WA9DIGDI1

Numeracy Digital literacy

Design and prototype the user experience (UX) of a digital system based on user requirements

WA9DIGDI2

Critical and creative thinking Digital literacy

Design algorithms that use functions and represent them as flow charts and/or pseudocode

WA9DIGDI3

Digital literacy

Predict the output of an algorithm using a given range of test cases and compare against actual output

WA9DIGDI4

Digital literacy

Implement, modify and debug programs that use functions in a general-purpose programming language

WA9DIGDI5

Digital literacy

Design thinking skills

Project management

Manage projects, using suitable technologies, with an agile and collaborative approach. Use project management processes to consider time, risk, economic and sustainable factors

WA9DIGDTPM1

Numeracy Critical and creative thinking Personal and social capability Ethical understanding

Investigating and defining

Ideate a problem and define the needs of an end user, through interviews and/or surveys

WA9DIGDTID1

Literacy Critical and creative thinking

Develop a design brief for a solution based on end user needs

WA9DIGDTID2

Critical and creative thinking

Investigate a range of technologies, resources and/or components to develop ideas and solutions, with consideration of social, ethical and other constraints

WA9DIGDTID3

Critical and creative thinking

Designing

Design alternative solutions considering available technologies, usability and aesthetics, using appropriate technical terms

WA9DIGDTDE1

Critical and creative thinking

Producing and implementing

Select, implement and test a range of technologies, techniques and processes to produce designed solutions and/or prototypes

WA9DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Evaluate design processes and solutions against student-developed criteria

WA9DIGDTEV1

Literacy Critical and creative thinking

Year 10

Year Level Description

In the middle adolescence phase of schooling, teaching and learning programs encourage students to develop an open and questioning view of themselves as active participants in their society and the world.

Digital Technologies further develops student understanding and skills in computational thinking, such as precisely and accurately describing problems, and the use of modular approaches to solutions. They investigate the potential and challenges of increased knowledge and choice of technologies; and an understanding of the relationship between knowledge, technologies and consumer and/or producer values.

In Year 10, students consider how human interaction with networked systems introduces complexities surrounding access to, and the security and privacy of, data of various types. They interrogate security practices and techniques used to compress data. Students explore the role of hardware and software in managing, controlling and securing access to data, in networked digital systems focusing on user or software supply chain vulnerabilities.

Students apply design thinking by using divergent techniques to generate design ideas for user experiences and solutions. They develop and represent documents online as content (text), structure (mark-up) and presentation (styling). Students analyse problems and design, and implement and evaluate a range of solutions, such as database-driven websites, artificial intelligence engines and simulations. They design and implement algorithms involving functions and logical operators and, where appropriate, represent them as flow charts. Students further develop project management skills through developing detailed plans with consideration of time, production processes, social, ethical, economic and sustainability factors, and legal responsibilities.

Achievement Standard

By the end of the year:

Students describe how hardware and software manage, control and secure access to data in networked digital systems with a focus on software supply chain vulnerabilities. They represent documents online as content (text), structure (mark-up) and presentation (styling) and explain the purpose of these distinctions. Students analyse and visualise data interactively using a range of software, including spreadsheets and relational databases, to draw conclusions and make predictions based on identified trends and outliers. They model and query entities and relationships using structured data. Students apply the Australian Privacy Principles to critique systems and manage the digital footprint of individuals.

Students define and decompose real-world problems and use data gathering techniques to create user experiences and user interfaces. They design and prototype the user experience of a digital system and algorithms involving functions, modules and logical operators, and represent them as flow charts and/or pseudocode. Students validate algorithms and programs by comparing their output against a range of test cases. They implement, modify and debug modular programs, applying algorithms and data structures, in a programming language.

In Digital Technologies, students identify the needs of the client or stakeholder to determine the basis for a solution. They develop and critique design briefs for a designed solution. Students investigate components and resources to develop increasingly sophisticated solutions, identifying and considering associated constraints. They apply design thinking, creativity, enterprise skills and innovation to develop, modify and communicate detailed design ideas. Students design possible solutions and analyse designs against criteria, including functionality, accessibility, usability and aesthetics, using appropriate technical terms and technologies. They select, justify, implement and test appropriate technologies and processes to produce designed

solutions. They manage projects, using digital technologies with an agile and collaborative approach, while considering time, production processes, social, ethical, economic and sustainable factors, and legal responsibilities.

Content Descriptions

Digital systems

Hardware and software are used to manage, control and secure access to data in networked digital systems

For example:

- public key cryptography, such as TLS, and hashing
- secure storage and transmission of data, such as SHA-1
- private information moves through a system and can be identified as the most likely target of a cyber attack. Data packets can be mapped when moving between the user and server in a web application. Sending data in plain text becomes susceptible to a 'man-in-the-middle' attack
- networks can be configured by using real or simulated hardware where data packets move around the network, with various levels of network efficiency
- data moves through a network based on layers of the TCP/IP model
- cybersecurity threat models are essential tools to identify and understand potential threats to a system, enabling the design and implementation of appropriate security measures
- a cybersecurity threat model should consider assets, threats, attack vectors, vulnerabilities, risk, security controls and the environment

WA10DIGDS1

Digital literacy

Data representation

Represent documents online as content (text), structure (mark-up) and presentation (styling) and the purpose of these distinctions

For example:

- documents are represented by separating content (the text in the document), structure (the document structure, such as headings and paragraphs) and presentation (document layout and style) in digital publications
- correct HTML tags to allow for accessibility, such as screen readers
- maintenance and updating of content and/or style is easier with correct document structure
- HTML (content data) and CSS (presentation data) are used in conjunction to create websites

WA10DIGDR1

Digital literacy

Acquiring, managing and analysing data

Analyse and visualise data interactively using a range of software, including spreadsheets and/or relational databases, to draw conclusions and make predictions by identifying trends and outliers

For example:

- interactive visualisations are used for exploring complex data, such as population, life expectancy and fertility rates in motion charts
- explore machine learning, a form of artificial intelligence (AI), where an algorithm is trained using a dataset, such as to classify images into categories

WA10DIGAD1

Numeracy Digital literacy

Model and query entities and their relationships using structured data

For example:

- modelling entities and processes, their attributes, and the relationships between them
- creating database tables for a movie, a user and their movie review, where a movie has a title, genre and release date, and a review has a movie, a user and their rating and comments
- interpreting and querying multi-table databases using SQL queries with SELECT, WHERE and simple JOIN/GROUP BY clauses and counting, such as checking that each teacher is only allocated to one class at a time

WA10DIGAD2

Numeracy Digital literacy

Privacy and security

Australian Privacy Principles (APP) are used to critique systems and manage the digital footprint of individuals

For example:

- the APPs are a set of 13 guidelines that form the foundation of Australia's privacy law, specifically the Privacy Act 1988. These principles govern how organisations handle personal information, ensuring it is collected, used, and managed in a way that protects individuals' privacy
- when using the APPs to critique systems such as websites, apps, or online services, individuals or regulators assess whether these systems comply with privacy laws. This involves examining how personal data is collected, stored, used, and shared; for example, a system can be critiqued based on transparency, data minimisation and security

WA10DIGPS1

Ethical understanding Digital literacy

User or software supply chain vulnerabilities

WA10DIGPS2

Literacy Digital literacy

Digital implementation

Define and decompose real-world problems by using data gathering techniques to create the client needs

WA10DIGDI1

Numeracy Digital literacy

Design and prototype the user experience and user interface (UX/UI) of a digital system based on client needs

WA10DIGDI2

Critical and creative thinking Digital literacy

Design modular algorithms involving functions and logical operators (AND, OR, NOT) and represent them as flow charts and/or pseudocode

WA10DIGDI3

Digital literacy

Validate algorithms and programs by comparing output against a range of test cases

WA10DIGDI4

Numeracy Digital literacy

Implement, modify and debug modular programs, applying algorithms and data structures in a general-purpose programming language

WA10DIGDI5

Digital literacy

Design thinking skills

Project management

Manage projects, using suitable technologies, with an agile and collaborative approach. Use project management processes to consider time, production processes, social, ethical, economic and sustainable factors, and legal responsibilities

WA10DIGDTPM1

Numeracy Critical and creative thinking Personal and social capability Ethical understanding

Investigating and defining

Ideate a problem and define the needs of the client/stakeholder through anecdotal evidence and/or data gathering techniques

WA10DIGDTID1

Literacy Critical and creative thinking

Develop a design brief for a solution or to innovate an existing product, service or environment

WA10DIGDTID2

Critical and creative thinking

Investigate a range of technologies, resources and/or components to develop ideas and solutions, with consideration of social and ethical factors, legal responsibilities and competing constraints

WA10DIGDTID3

Critical and creative thinking

Designing

Design alternative solutions considering available technologies, functionality, accessibility, usability and aesthetics, using appropriate technical terms

WA10DIGDTDE1

Critical and creative thinking

Producing and implementing

Select, justify, implement and test a range of technologies, techniques and processes to produce solutions and/or prototypes

WA10DIGDTPI1

Critical and creative thinking Personal and social capability Digital literacy

Evaluating

Evaluate design processes and solutions against student-developed criteria

WA10DIGDTEV1

Literacy Critical and creative thinking